Smart Portable Thermos

Project Overview

This project focuses on designing a smart portable thermos that allows users to prepare hot beverages and powdered soups during outdoor activities. The product integrates a heating system inside the thermos body and includes a simple digital interface for temperature and preparation control.

Users can prepare coffee, tea, or powdered soup, and adjust the soup density depending on their preference. The product also includes a hygiene indicator that shows when the thermos should be cleaned to maintain safe usage.

The goal is to create a portable, multifunctional, and ergonomic product that improves convenience during travel, camping, or daily commuting.

Keywords

- portability
- smart thermos
- outdoor beverage preparation
- powdered soup preparation
- adjustable density

- hygiene monitoring
- user convenience
- ergonomic design
- multifunctional product

Design Milestones / Process Timeline

1. Problem Identification
Many people want to consume hot drinks or quick meals during outdoor activities. However, existing thermoses only store drinks and cannot prepare them.

2. Research Phase
Existing thermoses and portable beverage devices were analyzed to understand usability, heating technologies, and portability challenges.

3. Concept Development
Different ideas were explored to combine a thermos with a heating and preparation system.

4. AI Assisted Ideation
AI tools were used to generate alternative product concepts and evaluate usability improvements.

5. Concept Selection
A portable smart thermos with an integrated heating system was selected as the main concept.

6. Feature Expansion
Additional functions were introduced, such as powdered soup preparation, density adjustment, and a hygiene monitoring system.

7. Design Refinement
The form, interface placement, and portability features were refined to improve ergonomics and usability.

8. Final Concept
The final design is a multifunctional smart thermos capable of heating water, preparing beverages, and mixing powdered soup while providing hygiene feedback to the user.

Selected Impactful Prompts

The product is designed as a portable thermos that can prepare coffee or tea outdoors.
How could this concept be expanded to support other instant foods such as powdered soup?
Please suggest features that would improve usability, such as adjusting the soup density or helping users maintain hygiene during use.

Based on the portable thermos concept, please suggest improvements related to ergonomics, interface design, and portability.
How can the display, controls, and carrying system be designed to make the product more practical for students, travelers, and outdoor users?

I am an industrial design student developing a portable smart thermos that can heat water and prepare beverages during outdoor activities.
Can you help me refine this concept and suggest how the heating system and brewing chamber could be integrated inside the thermos body while keeping the product compact and portable?

AI Role Tags

Role	Contribution
Clarifier	Helped clarify how heating and preparation systems could be integrated into the thermos body.
Re-framer	Suggested expanding the product from only coffee preparation to a multifunctional food and beverage device.
Expander	Generated ideas such as soup preparation, density adjustment, and hygiene monitoring features.
Critique	Highlighted potential usability improvements and suggested simplifying the interface.

AI Response Summaries

AI responses mainly helped to:

- improve the usability of the product
- expand the product functionality
- suggest ergonomic improvements
- refine the placement of internal components

These insights supported the development of a more practical and user-centered design.

Reflection

Using AI during the design process helped generate ideas quickly and explore different product possibilities. AI suggestions encouraged the expansion of the concept beyond a simple coffee device to a multifunctional thermos that can prepare both beverages and powdered soup.

However, the final design decisions were made by evaluating practicality, usability, and user needs.

Design Decision

Key design decisions include:

- Integrating the heating system inside the thermos body
- Keeping the interface compact and simple
- Allowing users to prepare both beverages and powdered soups
- Adding a density adjustment feature for soup preparation
- Including a hygiene indicator that informs users when cleaning is needed
- Designing the product for portability and outdoor use

Accepted, Rejected, and Transformed Ideas

Accepted

- Thermos-based portable design
- Integrated heating system
- Simple digital interface
- Powdered soup preparation feature
- Hygiene indicator

Rejected

- Large touchscreen interface
- Complex brewing mechanisms that increase cost

Transformed

- The initial concept of a portable coffee maker was transformed into a multifunctional smart thermos capable of preparing drinks and soup.

Thank You